Object Units or Pixels? A Negative Proxy Feasibility Study for Online Segmentation Adaptation

Anonymous Author(s)

Abstract

We study a pre-registered question about online open-vocabulary semantic segmentation adaptation: under a matched update budget, are CLIP-verified object units a better pseudo-supervision substrate than pixels or ungated masks? The intended CAT-Seg, Cityscapes, ACDC, and MLMP benchmark could not be executed in this workspace because those assets were unavailable, so we report an explicitly reframed proxy feasibility study instead. The executed protocol uses Pascal VOC 2012, a DeepLabV3-ResNet50 proxy backbone, a rank-4 logit adapter, six synthetic domain blocks, and a frozen calibration split that sets the acceptance thresholds and shared support budget once for all methods. Across the 144-image main proxy stream, CLIP_VERIFIED reaches 0.5386 ± 0.0083 foreground mIoU, while TOPB_PIXEL reaches 0.54333 ± 0.00002, RAW_MASK reaches 0.54332 ± 0.00001, and the frozen baseline reaches 0.54330. Paired Wilcoxon signed-rank tests show no significant advantage for CLIP_VERIFIED over TOPB_PIXEL ($p = 0.6334$ on order A, $p = 0.8145$ on order B) or over RAW_MASK ($p = 0.5060$ on order A, $p = 0.7301$ on order B). The result is therefore negative proxy evidence: in this executable closed-vocabulary setting, CLIP verification does not reliably improve online adaptation and instead adds variance and runtime.

1 Introduction

Online test-time adaptation for semantic segmentation is now well established in both closed-vocabulary and open-vocabulary settings [Das et al., 2023, Sojka et al., 2023, Noori et al., 2025, De Silva et al., 2025]. In parallel, recent open-vocabulary semantic segmentation (OVSS) systems show that mask-level or object-level reasoning is often useful at inference time [Liang et al., 2023, Xu et al., 2023, Shan et al., 2024, Lee et al., 2024, Kim et al., 2024]. What remains less clear is whether object-like units are the right *adaptation substrate* when the backbone, optimizer, and update budget are all held fixed.

The registered intent for this project was to answer that question on CAT-Seg with Cityscapes, Cityscapes-C, ACDC, and MLMP as contextual comparison. A workspace-wide compatibility check found none of those assets, so that benchmark was not runnable. Rather than presenting an unexecuted plan as evidence, we report only the study that was actually run: a Pascal VOC 2012 proxy benchmark with synthetic domain shifts, a shared one-step-per-image adaptation recipe, and four matched substrates.

The executed study compares four methods: a frozen baseline, top-confidence pixels (TOPB_PIXEL), ungated proposal regions (RAW_MASK), and CLIP-verified proposal regions (CLIP_VERIFIED). Calibration is performed once on a disjoint 24-image split, fixing $\tau_{\text{ovss}} = 1.0$, $\tau_{\text{clip}} = 1.0$, a margin threshold of 0.9996290683746338, an entropy threshold of 0.0020806991495192053, an agreement threshold of 0.9819277108433735, and a shared budget $B = 0.0283355712890625$ of image area.

Our contributions are:

Preprint. Under review.

- We present a fully executed proxy protocol that isolates the substrate question under matched adaptation settings when the registered OVSS benchmark is unavailable.
- We provide a calibrated comparison across pixels, ungated masks, and CLIP-verified masks over two six-domain stream orders with fixed seeds $\{13, 17, 23\}$.
- We report a negative result: CLIP-verified object units do not outperform the simpler proxy baselines and exhibit higher variance, larger transition drops, and similar or worse runtime.

Section 2 reviews prior work, Section 3 describes the proxy protocol, Section 4 presents the experiments, Section 5 discusses implications and limitations, and Section 6 concludes.

2 Related Work

Our study sits at the intersection of OVSS, test-time adaptation, and region-based supervision.

Open-vocabulary semantic segmentation. Recent OVSS models already establish strong mask-aware inference pipelines. Mask-adapted CLIP [Liang et al., 2023] and Side Adapter Network [Xu et al., 2023] showed that masked regions and side adapters are effective for open-vocabulary prediction. Later work continued this trend by improving image embedding balance [Shan et al., 2024], combining SAM with OVSS decoders [Lee et al., 2024, Chen et al., 2025], and distilling object-context structure [Kim et al., 2024]. These papers motivate comparing object-like support units, but they do not isolate whether such units are helpful for *online adaptation* under a shared update recipe.

Test-time and source-free adaptation. Online adaptation for segmentation has been studied for continual domain shifts [Das et al., 2023, Sojka et al., 2023, Lee et al., 2025]. In OVSS, recent work explores vision-language model adaptation [Noori et al., 2025], test-time optimization [De Silva et al., 2025], vocabulary alignment [Mazzucco et al., 2025], and semantic library adaptation [Qorbani et al., 2025]. These papers confirm that adaptation itself is viable, but they optimize broader pipelines rather than asking whether pixels, superpixels, raw masks, or verified masks are the best adaptation substrate once the rest of the scaffold is fixed.

Regionization and foundation models. Our proxy study uses proposal regions and a CLIP-based agreement check, following the broader observation that segmentation quality can benefit from high-quality region proposals and vision-language verification [Radford et al., 2021, Kirillov et al., 2023]. We also include a SLIC superpixel control on the reduced slice, motivated by classic regionization work [Achanta et al., 2012]. Unlike prior work, our goal is not to introduce a stronger architecture, but to audit whether richer support units help under matched online updates.

3 Method

3.1 Executed Scope

The registered CAT-Seg benchmark was infeasible in this workspace. The executed study is therefore a *proxy feasibility study* with a narrower claim: whether CLIP-verified object-like regions outperform matched proxy baselines under the same online adaptation scaffold. All conclusions in this paper are restricted to that proxy setting.

3.2 Proxy Benchmark

The proxy backbone is `torchvision` DeepLabV3-ResNet50 [Chen et al., 2017] operating on the Pascal VOC 2012 label space [Everingham et al., 2015]. The only trainable component is a rank-4 logit adapter, following the same low-rank-update intent across all adaptive methods [Hu et al., 2022]. Inputs are resized to 256×256, batch size is 1, and each image receives exactly one online update step.

We evaluate six domains: clean, fog, gaussian noise, snow, dusk, and night. The calibration split contains 24 clean images. The main benchmark contains 144 validation images with 24 images per

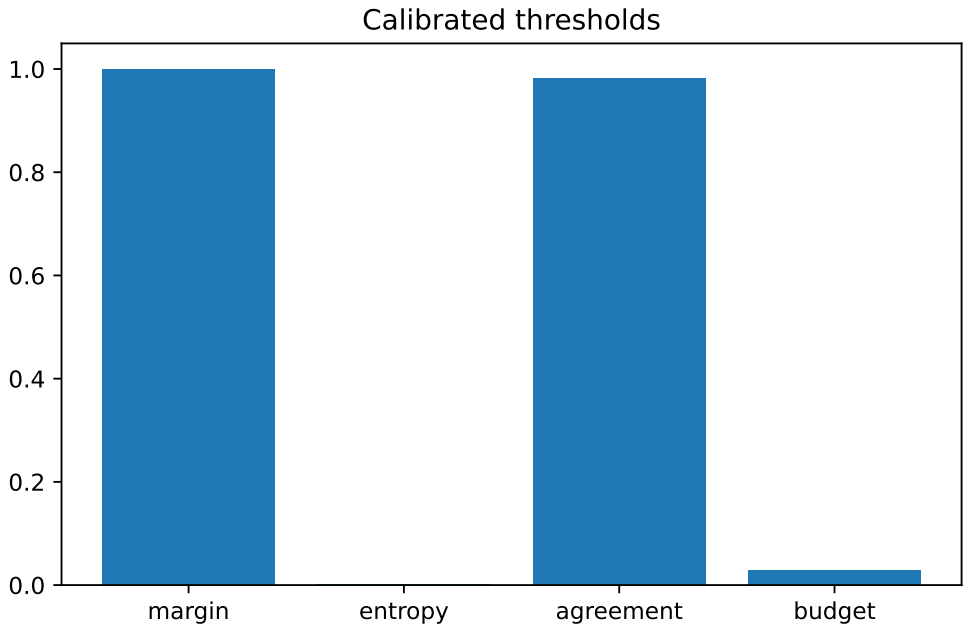

Figure 1: Calibration diagnostics on the 24-image held-out split. The thresholds used in all later runs are frozen after this stage: margin 0.9996290683746338, entropy 0.0020806991495192053, agreement 0.9819277108433735, and budget $B = 0.0283355712890625$.

domain. The reduced ablation slice contains 72 images with 12 per domain. Two stream orders are used:

- Order A: clean $\rightarrow$ fog $\rightarrow$ gaussian_noise $\rightarrow$ snow $\rightarrow$ dusk $\rightarrow$ night.
- Order B: night $\rightarrow$ dusk $\rightarrow$ snow $\rightarrow$ gaussian_noise $\rightarrow$ fog $\rightarrow$ clean.

3.3 Compared Substrates

The methods differ only in how they choose support for the single online update:

- **Frozen**: no adaptation.
- **TOPB_PIXEL**: use the top-confidence pixels up to the shared area budget B.
- **RAW_MASK**: use cached proposal regions without CLIP verification.
- **CLIP_VERIFIED**: use cached proposal regions that satisfy CLIP-agreement checks.

This matched setup is the central design choice: backbone, optimizer, trainable parameterization, one-step schedule, and budget accounting remain fixed, while the support substrate changes.

3.4 Calibration

Thresholds are frozen from the calibration split and never fit on the target stream. The executed calibration produces $\tau_{\text{ovss}} = 1.0$, $\tau_{\text{clip}} = 1.0$, margin threshold 0.9996290683746338, entropy threshold 0.0020806991495192053, agreement threshold 0.9819277108433735, and shared budget $B = 0.0283355712890625$. Among 323 eligible masks, the calibration means are 0.9159 for the fused margin, 0.1067 for entropy, and 0.7935 for agreement. The accepted-area mean is 0.0299 over the 24 calibration images. Calibration also records 25 rejections from disjoint top-3 predictions and 36 rejections where the fused label is not in both top-3 sets.

Table 1: Main proxy comparison on the 144-image benchmark. Metrics are aggregated exactly from `results.json`. Higher is better for mIoU and lower is better for runtime and transition drop.

Method	Foreground mIoU ↑	Runtime / image (s) ↓	Skip ratio ↓	Avg. transition drop ↓
Frozen	0.54330	0.00753	0.000	0.00000
TOPB_PIXEL	**0.54333 ± 0.00002**	0.09047	**0.000**	0.00001
RAW_MASK	0.54332 ± 0.00001	**0.05529**	0.875	**0.00000**
CLIP_VERIFIED	0.53859 ± 0.00830	0.09120	0.374	0.01386

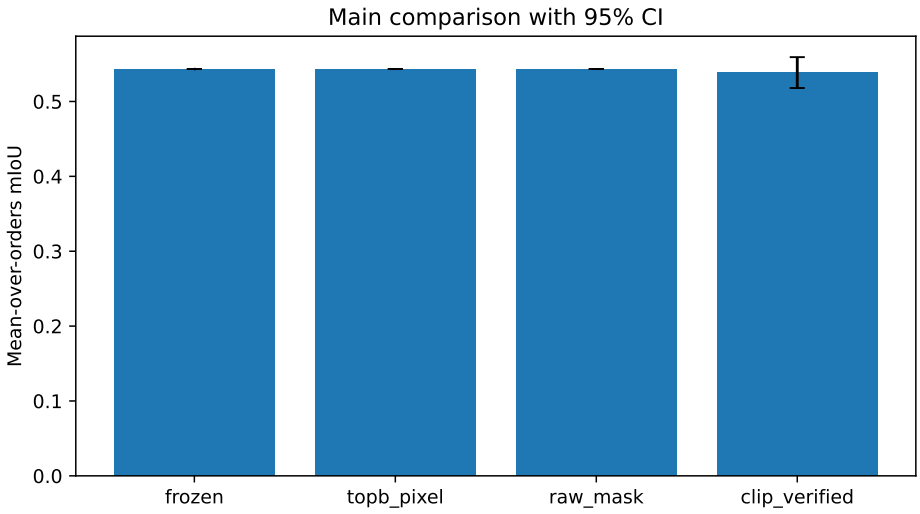

Figure 2: Main comparison with error bars over seeds $\{13, 17, 23\}$. The object-unit method CLIP_VERIFIED has the lowest mean and by far the largest variance among the adaptive methods.

3.5 Metrics

The primary metric is dataset-level foreground mIoU computed from the confusion matrix over the 20 non-background Pascal VOC classes. Per-image mIoU is computed only over foreground classes present in that image. Older zero-filled per-image diagnostics are retained only as diagnostics and are not used for aggregate reporting.

4 Experiments

4.1 Setup

All runs use Python 3.10.12 on a machine with one NVIDIA RTX A6000 (49140 MiB) and 4 CPU cores. Cache construction confirms 168 proposal caches, 168 SLIC caches, and one CLIP text-feature cache for the unique benchmark samples. A dry run on 12 images estimates 0.14956569656108817 projected GPU hours for the full executed matrix, with per-image timing components of 0.007636378674457471 seconds for inference, 0.046041400637477636 seconds for support selection, and 0.09065962902137212 seconds for the update step. Adaptive methods use seeds 13, 17, and 23; the frozen baseline is deterministic and run once per order.

4.2 Main Results

Table 1 and Figure 2 summarize the main comparison. The central outcome is negative: CLIP_VERIFIED does not improve over the simpler substrates. Its mean foreground mIoU is 0.5385898991422345, substantially below TOPB_PIXEL at 0.5433288613242264, RAW_MASK at 0.5433192550427369, and even the frozen baseline at 0.543300277035063.

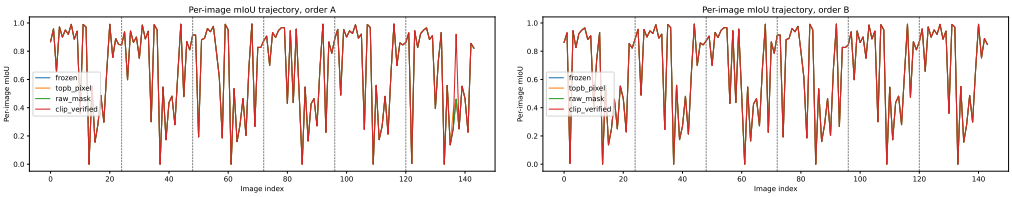

Figure 3: Per-domain trajectories for the two stream orders. The order reversal changes domain exposure, but it does not change the qualitative conclusion: the verified-region method does not open a stable margin over the matched baselines.

Table 2: Reduced-slice ablations on order A. Values are aggregated over seeds $\{13, 17, 23\}$. The calibrated CLIP_VERIFIED setting is `threshold_+0.00` and `clip_verified_budget_1.0`.

Ablation	Foreground mIoU ↑	Runtime / image (s) ↓	Skip ratio ↓
CLIP_VERIFIED ($1.0B$)	0.46884 ± 0.00002	0.09108	0.319
CLIP_VERIFIED ($0.8B$)	0.46884 ± 0.00002	0.08606	0.403
CLIP_VERIFIED ($1.2B$)	0.46884 ± 0.00003	0.09236	0.306
CLIP_VERIFIED (-0.05 margin offset)	0.46885 ± 0.00002	0.09071	0.315
CLIP_VERIFIED ($+0.05$ margin offset)	0.46884 ± 0.00001	0.04751	1.000
No CLIP	0.46884 ± 0.00001	0.05250	0.917
No consistency	0.46885 ± 0.00002	0.09219	0.319
SLIC	0.46884 ± 0.00001	**0.03957**	1.000
TOPB_PIXEL ($1.0B$)	**0.46886 ± 0.00002**	0.09009	**0.000**

The per-domain means tell the same story. For CLIP_VERIFIED, the clean-domain mean is only 0.5443230588407804, versus 0.5539553930468121 for TOPB_PIXEL, 0.5539649992975086 for RAW_MASK, and 0.5539636611350314 for Frozen. Across fog, gaussian noise, snow, dusk, and night, CLIP_VERIFIED remains close to the baselines but never recovers the clean-domain deficit enough to offset that loss.

Statistical testing also fails to support the original hypothesis. Paired Wilcoxon signed-rank tests over the paired per-image trajectories give $p = 0.6334355322075388$ and $p = 0.814461435845842$ for CLIP_VERIFIED versus TOPB_PIXEL on orders A and B, and $p = 0.5060364728194691$ and $p = 0.7301047145288457$ for CLIP_VERIFIED versus RAW_MASK. In this proxy benchmark, CLIP verification does not produce a statistically credible gain.

4.3 Ablations

The reduced 72-image slice probes whether the negative result is caused by a poor threshold or budget choice. Table 2, Figure 4, and Figure 5 show that the answer is no.

Three observations are consistent across the ablations. First, changing the verified-mask budget from $0.8B$ to $1.2B$ barely changes mIoU: the means are 0.46883589864144054, 0.468844746782759, and 0.46884258689505803. Second, relaxing the threshold by -0.05 or tightening it by $+0.05$ also has negligible effect on accuracy, while the strict $+0.05$ setting collapses into total skipping with skip ratio 1.0. Third, removing CLIP verification (`no_clip`) or removing the consistency term (`no_consistency`) leaves the reduced-slice result essentially unchanged. The simplest interpretation is that, in this proxy implementation, the support substrate is not the performance bottleneck.

5 Discussion and Limitations

The most important result is methodological: a controlled comparison can still be informative when it yields a negative answer. In the executed proxy benchmark, CLIP verification does not improve online adaptation relative to confidence-ranked pixels or ungated masks. It instead introduces larger variance (0.00830142308439634 versus $2.2690363616157517 \times 10^{-5}$ for TOPB_PIXEL), a larger average transition drop (0.013863096178038967 versus $1.2876245677018288 \times 10^{-5}$), and runtime comparable to the pixel baseline.

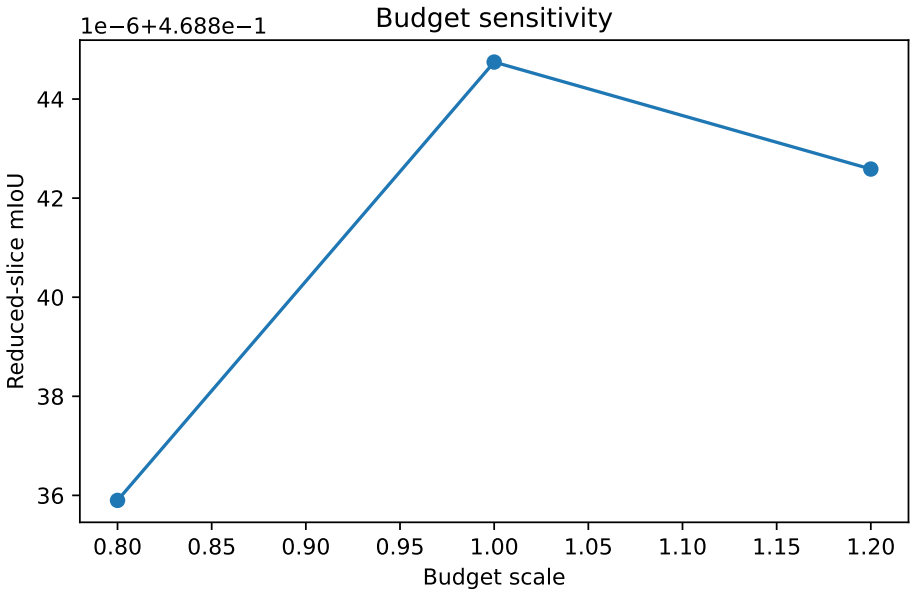

Figure 4: Budget sensitivity on the reduced slice. The verified-mask variant remains flat from $0.8B$ to $1.2B$, while TOPB_PIXEL remains slightly better throughout.

This does *not* refute the original OVSS hypothesis. The study is limited in several ways. First, it is a closed-vocabulary proxy on Pascal VOC rather than the intended CAT-Seg benchmark on Cityscapes, Cityscapes-C, and ACDC. Second, MLMP was not run because the required assets were unavailable, so the comparison is internal rather than benchmark-complete. Third, the reduced-slice SLIC result should be read only as an ablation control, because SLIC was not executed on the full 144-image matrix. Fourth, the adaptation scaffold is intentionally simple: one step per image and a rank-4 logit adapter may underuse the potential of object-level support.

Even with those limitations, the negative result is useful. It suggests that simply replacing pixels with verified regions is not enough, by itself, to obtain a robust advantage under strict budget matching. Future work should revisit the question only in the originally intended OVSS setting, or test whether object units become helpful when paired with stronger objectives, richer memory, or adaptation rules that explicitly model domain transitions.

6 Conclusion

We set out to test whether CLIP-verified object units are a better online adaptation substrate than pixels or ungated masks under matched budgets. The registered OVSS benchmark was unavailable, so we executed and report only a Pascal VOC proxy feasibility study with fixed calibration, two stream orders, and matched update rules.

The resulting evidence is negative. On the main 144-image proxy stream, CLIP_VERIFIED achieves 0.5385898991422345 foreground mIoU, below TOPB_PIXEL at 0.5433288613242264, RAW_MASK at 0.5433192550427369, and Frozen at 0.543300277035063, with no significant paired advantage over either adaptive baseline. Reduced-slice ablations remain flat across threshold and budget changes. Within this executable proxy setting, CLIP verification does not provide a reliable adaptation benefit.

The broader implication is simple: object-like support units should not be assumed to help online adaptation merely because they help inference-time segmentation. The substrate question remains open for the originally intended OVSS benchmark, but this workspace supports only the narrower conclusion reported here.

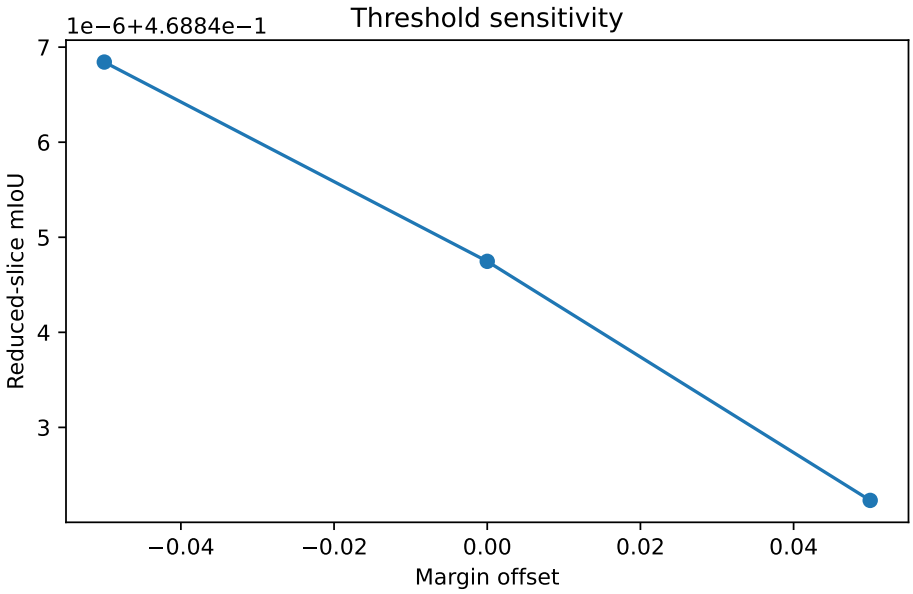

Figure 5: Threshold sensitivity for CLIP_VERIFIED. Accuracy is nearly unchanged across the tested offsets, and the strictest threshold drives the method into full skipping rather than improved precision.

References

Radhakrishna Achanta, Appu Shaji, Kevin Smith, Aurelien Lucchi, Pascal Fua, and Sabine Süsstrunk. SLIC superpixels compared to state-of-the-art superpixel methods. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(11):2274–2282, 2012.

Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam. Rethinking atrous convolution for semantic image segmentation. *arXiv preprint arXiv:1706.05587*, 2017.

Lin Chen, Yingjian Zhu, Qi Yang, Xin Niu, Kun Ding, and Shiming Xiang. SAM-MI: A mask-injected framework for enhancing open-vocabulary semantic segmentation with SAM. *arXiv preprint arXiv:2511.20027*, 2025.

Debasmit Das, Shubhankar Borse, Hyojin Park, Kambiz Azarian, Hong Cai, Risheek Garrepalli, and Fatih Porikli. TransAdapt: A transformative framework for online test time adaptive semantic segmentation. In *ICASSP*, 2023.

Ulindu De Silva, Didula Samaraweera, Sasini Wanigathunga, Kavindu Kariyawasam, Kanchana Ranasinghe, Muzammal Naseer, and Ranga Rodrigo. Test-time optimization for domain adaptive open vocabulary segmentation. *arXiv preprint arXiv:2501.04696*, 2025.

Mark Everingham, S. M. Ali Eslami, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The Pascal Visual Object Classes Challenge: A retrospective. *International Journal of Computer Vision*, 111(1):98–136, 2015.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022.

Chanyoung Kim, Dayun Ju, Woojung Han, Ming-Hsuan Yang, and Seong Jae Hwang. Distilling spectral graph for object-context aware open-vocabulary semantic segmentation. *arXiv preprint arXiv:2411.17150*, 2024.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *ICCV*, 2023.

Minhyeok Lee, Suhwan Cho, Jungho Lee, Sunghun Yang, Heeseung Choi, Ig-Jae Kim, and Sangyoun Lee. Effective SAM combination for open-vocabulary semantic segmentation. *arXiv preprint arXiv:2411.14723*, 2024.

Seunghwan Lee, Inyoung Jung, Hojoon Lee, Eunil Park, and Sungeun Hong. Instance-aware test-time segmentation for continual domain shifts. *arXiv preprint arXiv:2512.08569*, 2025.

Feng Liang, Bichen Wu, Xiaoliang Dai, Kunpeng Li, Yinan Zhao, Hang Zhang, Peizhao Zhang, Peter Vajda, and Diana Marculescu. Open-vocabulary semantic segmentation with mask-adapted CLIP. In *CVPR*, 2023.

Silvio Mazzucco, Carl Persson, Mattia Segu, Pier Luigi Dovesi, Federico Tombari, Luc Van Gool, and Matteo Poggi. Lost in translation? Vocabulary alignment for source-free adaptation in open-vocabulary semantic segmentation. In *BMVC*, 2025.

Mehrdad Noori, David Osowiechi, Gustavo Adolfo Vargas Hakim, Ali Bahri, Moslem Yazdanpanah, Sahar Dastani, Farzad Beizaee, Ismail Ben Ayed, and Christian Desrosiers. Test-time adaptation of vision-language models for open-vocabulary semantic segmentation. *arXiv preprint arXiv:2505.21844*, 2025.

Reza Qorbani, Gianluca Villani, Theodoros Panagiotakopoulos, Marc Botet Colomer, Linus Harenstam-Nielsen, Mattia Segu, Pier Luigi Dovesi, Jussi Karlgren, Daniel Cremers, and Matteo Poggi. Semantic library adaptation: LoRA retrieval and fusion for open-vocabulary semantic segmentation. In *CVPR*, 2025.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

Xiangheng Shan, Dongyue Wu, Guilin Zhu, Yuanjie Shao, Nong Sang, and Changxin Gao. Open-vocabulary semantic segmentation with image embedding balancing. In *CVPR*, 2024.

Damian Sojka, Yuyang Liu, Dipam Goswami, Sebastian Cygert, Bartlomiej Twardowski, and Joost van de Weijer. Technical report for ICCV 2023 visual continual learning challenge: continuous test-time adaptation for semantic segmentation. *arXiv preprint arXiv:2310.13533*, 2023.

Mengde Xu, Zheng Zhang, Fangyun Wei, Han Hu, and Xiang Bai. Side adapter network for open-vocabulary semantic segmentation. In *CVPR*, 2023.